

Jorge Luis Vargas Lara

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EDUCATION

University of California, Davis

Davis, CA

M.S. Water Resources Engineering, Dec. 2025 — B.S. Civil & Environmental Engineering, June 2024

- Johannes DeVries Graduate Student Award, 2025

TECHNICAL SKILLS

Programming: Python, Bash, MATLAB — **Modeling/GIS:** MODFLOW 6, FloPy, OpenFOAM, QGIS

Data: Pandas, NumPy, Matplotlib, GeoPandas, Excel — **Tools:** Git, GitHub, Linux, Jupyter, Astro (Web Development)

EXPERIENCE

Dauids Engineering

Davis, CA

Intern I

Sept. 2025 – Present

- **Remote Sensor and Weather Data Automation** | *Python, APIs, Pandas*
Analyzed weather station API documentation to design and implement automated data retrieval workflows for agricultural study plots, supporting multi-project analysis.
- **Water Diverter Analysis for SGMA Reporting** | *Python, QGIS*
Evaluated Tehama County water diverter datasets and mapped spatial usage patterns in QGIS to determine inclusion of additional data in SGMA annual reports.
- **ETo Data Evaluation for Tehama Integrated Hydrological Model** | *Python, QGIS, CIMIS*
Assessed external ETo datasets as alternative model inputs for Tehama IHM, documenting discrepancies and recommending sources to enhance SGMA reporting.
- **Cause of Decreasing Groundwater Usage in Annual SGMA Report – Tehama County** | *Word, QGIS*
Identified drivers of declining groundwater usage between yearly SGMA reports by analyzing historical diversion data; delivered an engineering-reviewed technical memo validating trends and standardizing data sourcing practices.

Morales Lab – UC Davis Civil and Environmental Engineering

Davis, CA

Graduate Researcher

July 2023 – Present

- **Biofilm Flow and Transport Modeling** | *OpenFOAM, Python*
Developed pore-scale CFD models to study biofilm impacts on flow, transport, and reactions in porous media, supporting groundwater remediation.
- **Image-Based Simulation Workflow** | *Python, Image Processing*
Designed and validated an image-processing workflow to convert segmented microscopy data into high-quality STL meshes for pore-scale transport simulations.
- **Simulation Automation and Post-Processing** | *Python, Bash, Linux*
Developed reproducible OpenFOAM pipelines that automated setup and post-processing, reducing manual effort.
- **Multiphase Oxygen Transport in Porous Media** | *Bash, Python*
Extended prior biofilm modeling workflows to simulate oxygen diffusion in porous systems containing gas bubbles.

Valley Roofing

Yucaipa, CA

Roofing Assistant – Seasonal

June 2019 – Present

- **Residential Roofing Fieldwork** Performed physically demanding fieldwork in high-temperature and elevated environments while following safety protocols.

Memorial Union, University of California, Davis

Davis, CA

Building Supervisor

Sep 2021 – Dec 2023

- **Client Facing Costumer Service** Led front desk operations, delivering customer service, overseeing public spaces, and coordinating staff operations

INDEPENDENT PROJECTS

Groundwater Flow and Transport Model – Colton, CA | *MODFLOW 6, FloPy, QGIS*

May 2025

- Developed a groundwater flow and transport model, integrating monitoring wells, topography, plume extents, and geospatial data in QGIS to evaluate aquifer behavior under realistic boundary conditions.

Guest Lecturer – Introduction to Turbulence | *Teaching, CFD, Documentation*

Dec 2025

- Designed and taught introductory CFD lessons, developing hands-on examples and mentoring students through term projects to independently build and validate simulations

KEY PUBLICATIONS

- Flow, Mixing, and Reaction Dynamics in Progressively Bioclogged Porous Media - Paper in preparation
- Sensitivity Analysis of Biofilm Permeability in Progressively Bioclogged Porous Media- Paper in preparation

LEADERSHIP

SCIENCE IS ART - CEE ART GALLERY

Founder and organizer of the first annual CEE Art Show, showcasing diverse research within the Civil and Environmental Engineering Department.

RELEVANT COURSEWORK

Turbulence - Water Resources Simulation - Soil Mechanics - Contaminant Fate and Transport in Porous Medium - Groundwater Systems Design - Sediment Transport - Hydrological Time Series Analysis - Deterministic Optimization and Design - Water Resource System Engineering- Jon Herman - Groundwater Modeling- Thomas Harter